

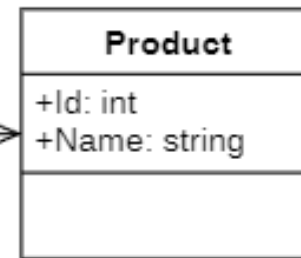
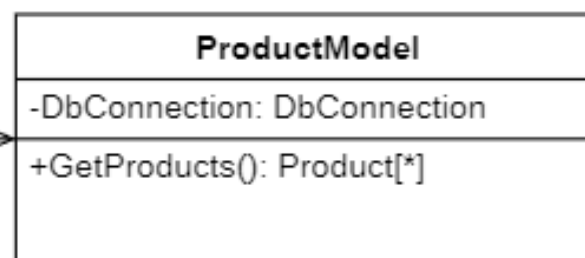
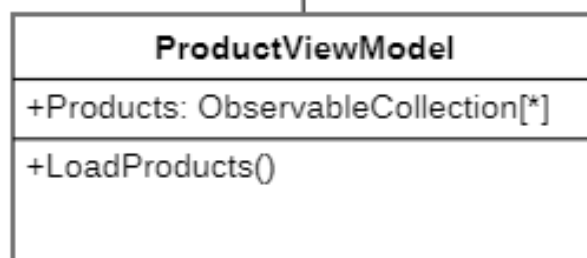
Software engineering

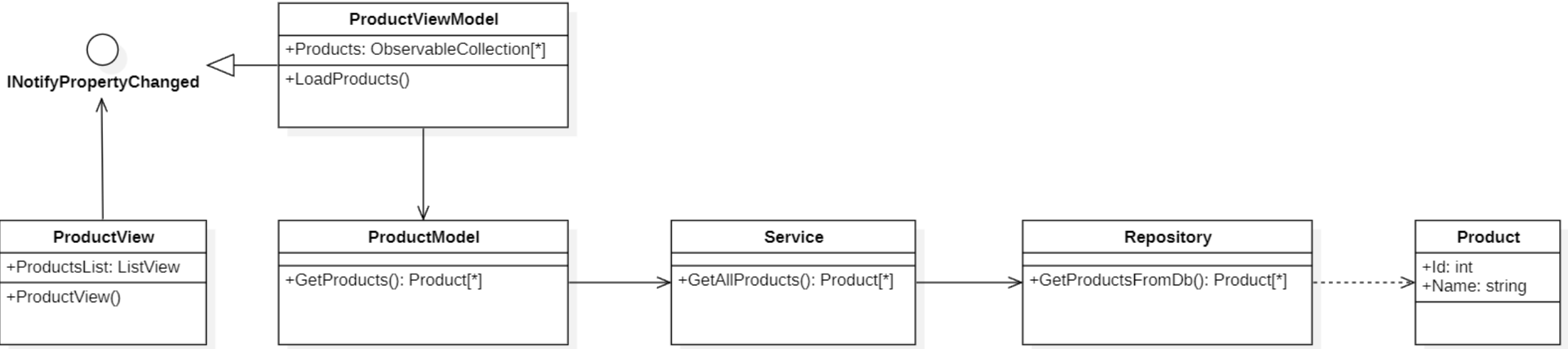
Getting hired

By applying this material, you should get:

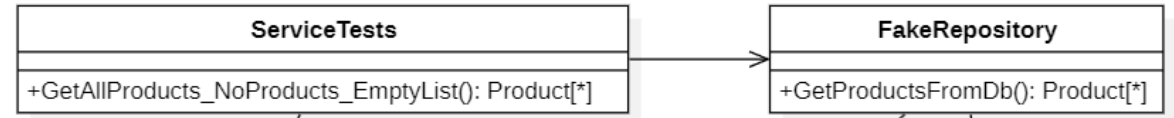
Money 4-6

INotifyPropertyChanged

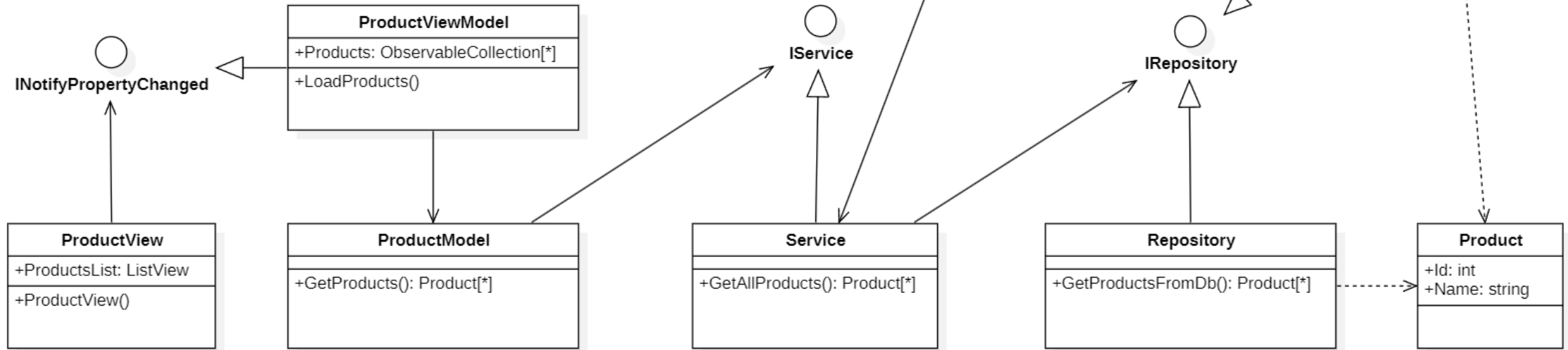




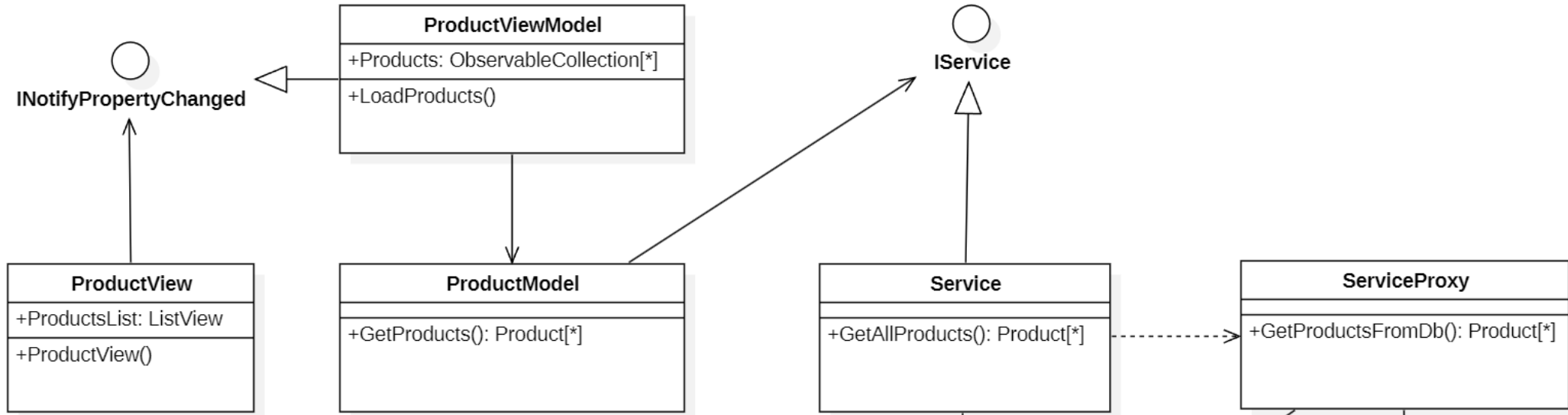
Testing project



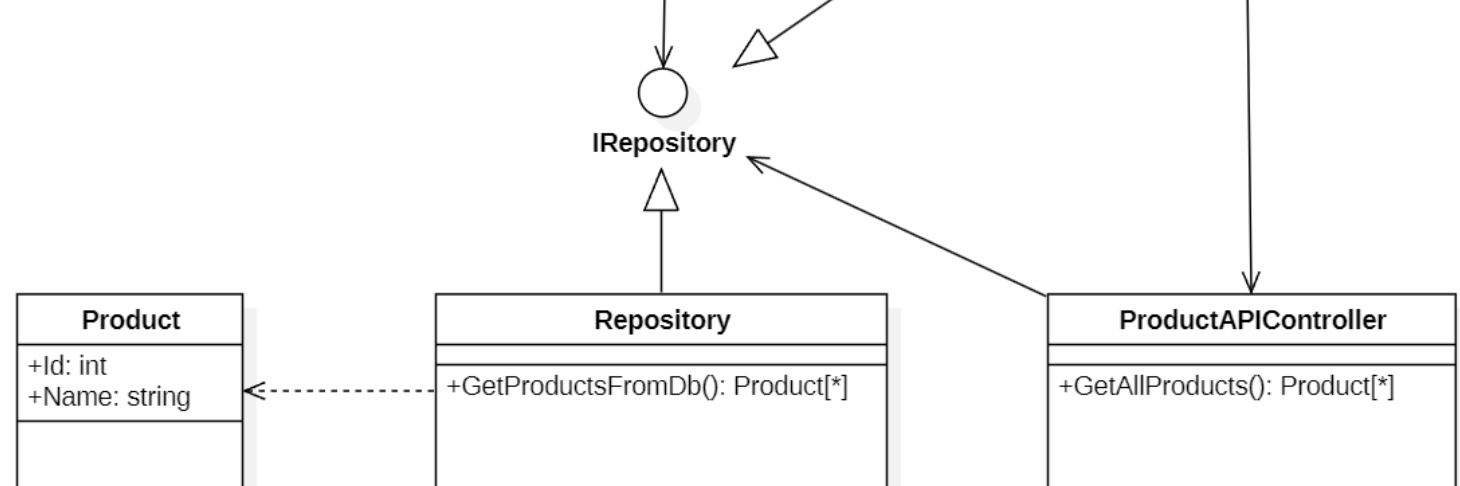
Application project

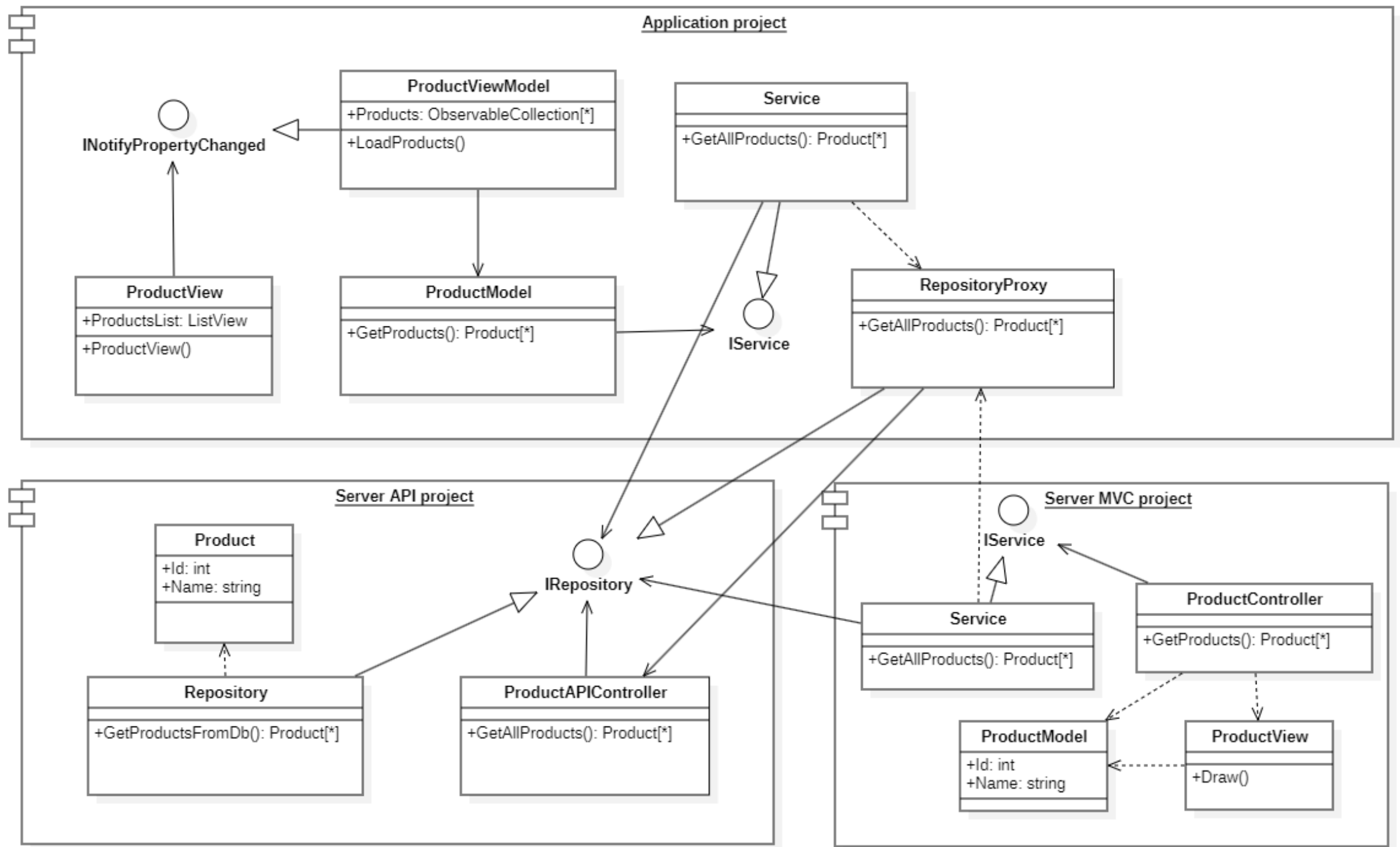


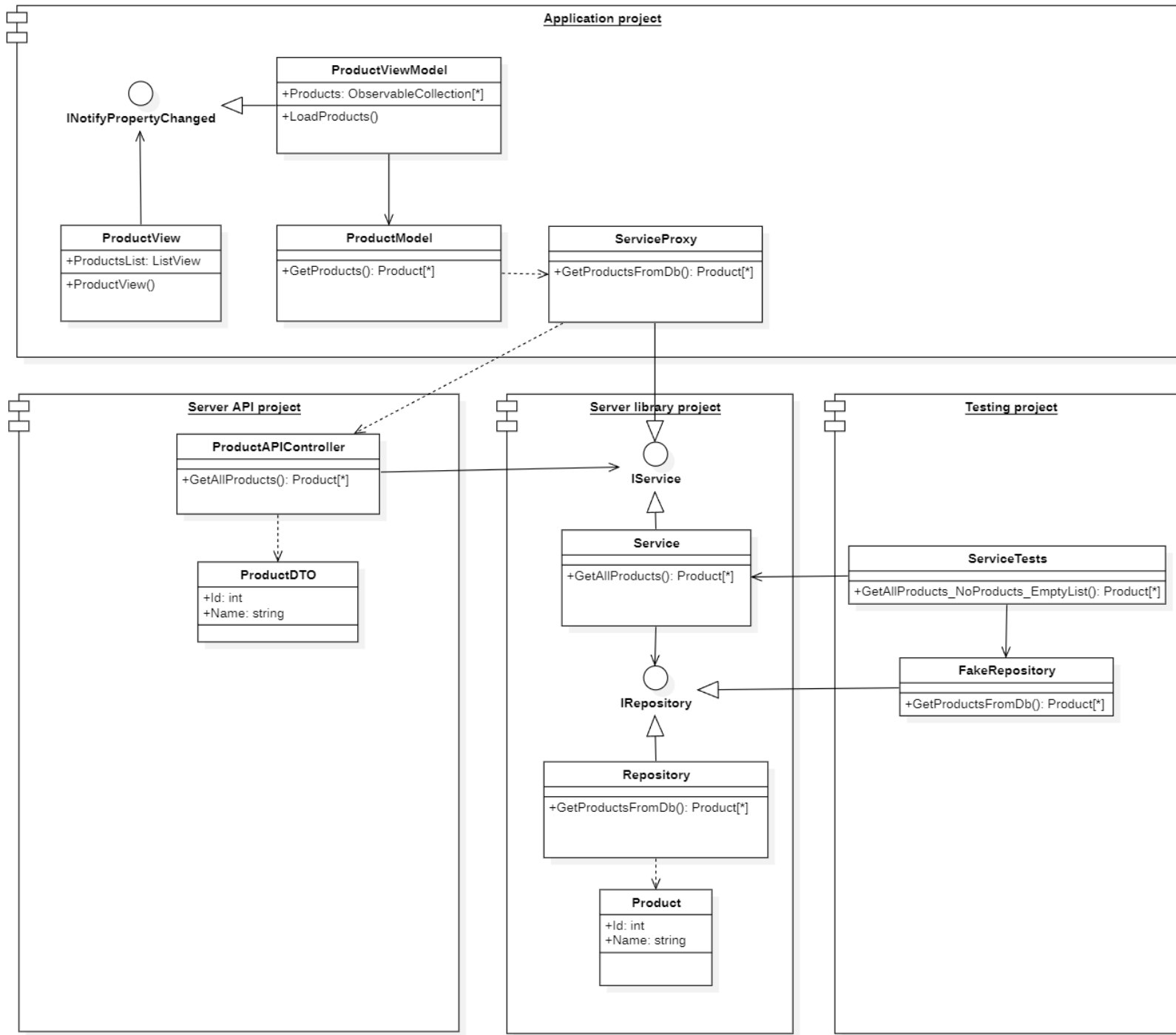
Application project

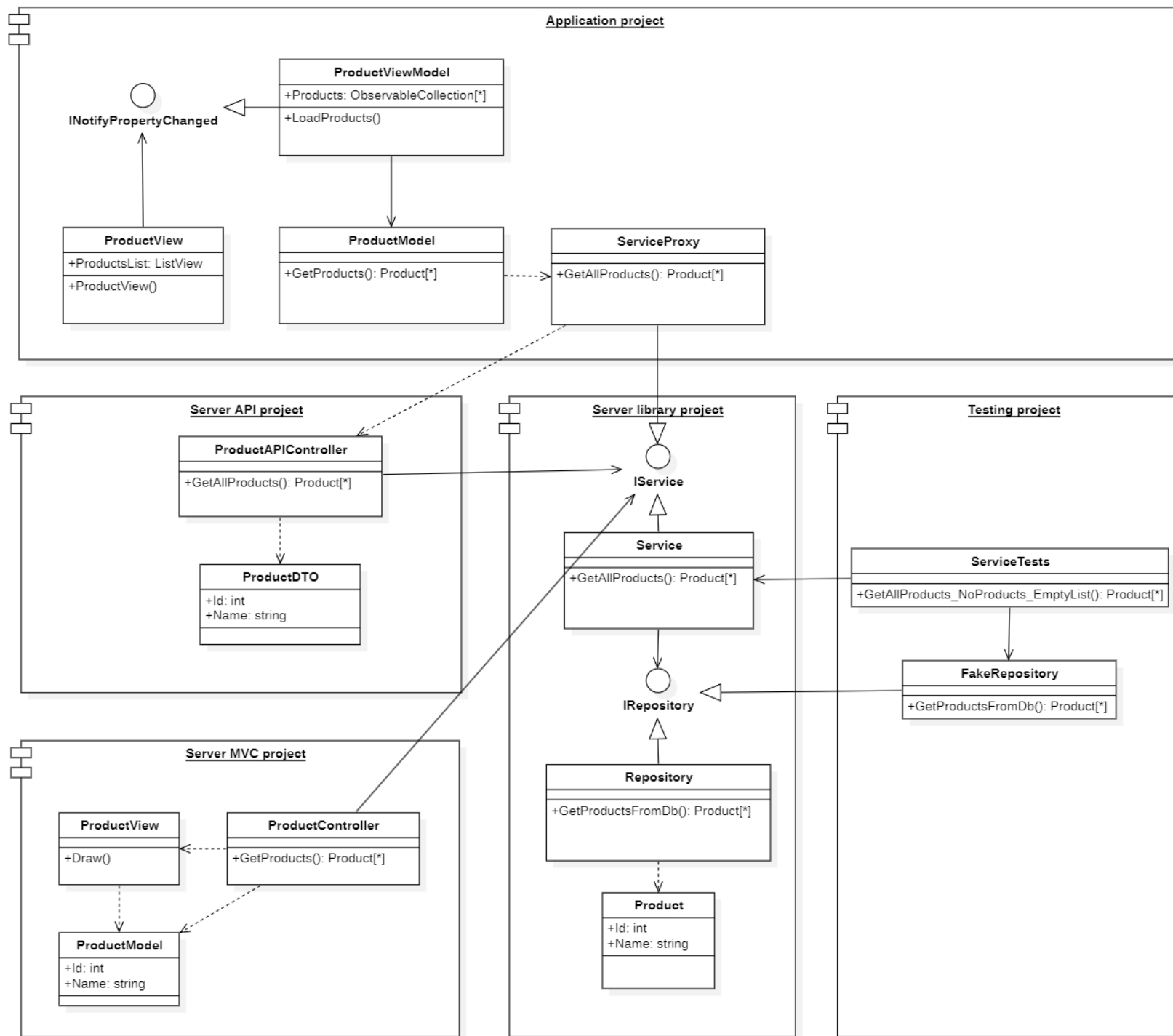


Server API project











C:\Users\Custos\source\repos X



1. Show all products
2. Add new product
3. Update product
4. Delete product
0. Exit

|

Logistics for product launch event

- The product launch event will take place at the last course (2nd of June 14:00) and the Saturday after (7th of June 9:00), both in 2/I.
- Presentations are done in order of labs, with the first 3 teams scheduled for Monday, if they run out of time then they will join the other teams Saturday.
- The team presenting will receive a +0.01 bonus to their final grade for each applauding non-team-member colleagues in the room when they finished their presentation.
- While not required at all, dressing according to project or any type of shenanigans short of pyrotechnics is encouraged.
- Grades for the launch event will be given by a panel of teaching assistants.
- The launch event is 10% of the grade.

What you should present

- Talk about your experience getting to this point
- Present what the project is about, and how much (percentage) of the initial requirements survived to the end
- Present your architecture
- Show your desktop and deployed web applications, demonstrating that changes in one are reflected in the other.
- The entire team will be on / around the podium, any number of team members may present, make sure that you have a HDMI port on the presenting laptop, and that it has a charger. Also make sure that you have plenty of data, and that you have tested everything you are going to present.

Picking a company







Hi, I'm _____!
Mile _____

Age: 4

Sex: male

Breed: Labrador Retriever

Size: 70 lbs

About Me

Milo is a very friendly labrador retriever who I adopted as a puppy. We've lived together in an apartment for the last 4 years and he is house-broken, quiet, and well-behaved. He's also received professional training and gets along very well with other animals, children, and adults.

Health

Neutered

Up to date on all vaccinations including

flea, tick, and worm prevention.

Obedience & Skills

Received a certificate of completion from obedience school. Gets along well with other animals and is kid-friendly.

CV

References

Joe Sanders | Previous landlord | 619.123.4567

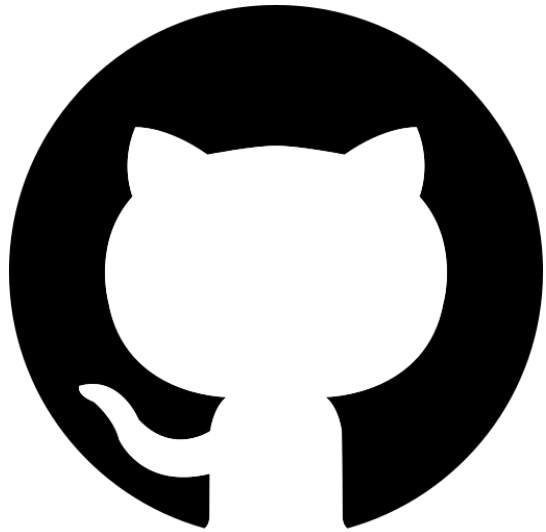
Jones | Veterinarian | 619.321.0987



What to put into yours?

- Requirements engineering +1-2
- UML +2
- Working in teams +2
- Git +3
- GUI/Design +2
- Software architectures +4
- C# + .NET +4
- Code quality +1-4
- Communication skill +1
- Unit testing +3-5
- Database access +1
- Web development +0-3
- Design patterns +3-4
- Scrum + Kanban +0-3
- DevOps +2

Use everything you have



stackoverflow



Project
Euler



Objective section



Experienced C# developer with more than five years of experience and a deep knowledge of OOP,



.NET Framework, .NET Core, Entity Framework, SQL, and web services development



Known for working effectively in a team and adhering to established schedules and deadlines



Constantly striving for professional growth, learning new technologies, and adapting to rapidly changing technical conditions



Try to highlight:
Experience, Technical proficiencies, Soft skills, Adaptability and lifelong learning

About Me section

- As a passionate technology enthusiast, my interest in coding has shaped my career as a C# Developer. I am known for my dedication to teamwork, meticulous attention to detail, and a knack for problem-solving. A firm believer in the power of continual learning, I'm always on the lookout for new technology and opportunities to further enhance my skills. My commitment to knowledge growth extends beyond my professional life. As an avid guitar player, I have learned the values of discipline, patience, and the pure joy of mastering a new piece. This hobby serves as a creative outlet, providing a balance to my technical pursuits.*
- Additionally, I enjoy contributing to open source projects, an experience that not only broadens my coding skills but also deepens my appreciation for the collaborative nature of our field. Integrity, collaboration, and the pursuit of excellence are my guiding principles, influencing both my personal and professional life. I am a hard-working individual with a unique blend of technical skills and creative flair, eager to make a significant impact in my future role.*

Tell me about a time when you had to solve a complex coding problem. How did you do it?

- **Situation:** I was working on a project to develop a new algorithm for image recognition. The algorithm was working well for most images, but it was failing for a small number of images. I needed to figure out why the algorithm was failing and fix the problem.
- **Task:** My task was to identify the bug in the algorithm and fix it.
- **Action:** I started by debugging the code. I looked for errors in the logic and the implementation. I also ran the algorithm on a variety of test images to see if I could reproduce the problem. After debugging the code, I still couldn't find the bug. I decided to try a different approach: I used a debugger to step through the code line by line. This allowed me to see exactly what the code was doing and where it was failing. Finally, I found the bug. It was a small mistake in the logic of the algorithm. I fixed the problem, and the algorithm started working correctly.
- **Result:** I was able to solve the problem, and the algorithm was working correctly for all of the images. I learned a lot from this experience, and I improved my debugging skills.

Questions during interviews HR

- Tell me about yourself.
- Why are you interested in this position? / Why do you want to work here?
- What are your greatest strengths? Weaknesses?
- What are your salary expectations?

Technical questions during interviews:

What are the principles of OOP in C#?

- **Encapsulation** allows data and methods to be bundled into a unit (class) and restricts access to certain components.
- **Inheritance** permits one class (the child or derived class) to inherit the attributes and methods of another class (the parent or base class). This promotes the reuse of code and establishes a hierarchical relationship between classes.
- **Polymorphism** is the capability of a single function or method to work in various ways based on its inputs or on which object it is called upon. In C#, polymorphism can be achieved through method overriding, using the `override` keyword, and method hiding, utilizing the `new` keyword to hide a method in the base class.
- **Abstraction** allows developers to hide complex implementations and show only the essential features of an object. This means that the user interacts with only what's necessary and the internal workings are kept hidden. In C#, abstract classes and interfaces are tools that can help achieve abstraction.

What does the dependency injection principle mean and how is it implemented in C#?

- **Dependency injection (DI)** is a software design approach that reduces tight coupling between system components. At the core of DI is the passing of dependencies (services, objects) to components rather than creating them inside those components. In C# and .NET, DI is often implemented using dependency containers, such as *DependencyInjection*, *Ninject*, *Autofac*, and others.

What are boxing and unboxing in C#, and why can they be a problem?

- **Boxing** is the process of converting a value type (e.g., *int*) to an object type or any interface type implemented by that value type. **Unboxing** is the reverse process, where a value from an object type is converted back to the corresponding value type. The primary concern with boxing and unboxing in C# is their potential to degrade application performance. Boxing necessitates heap memory allocation and value type copying, thus slowing operations, especially with large datasets or in high-frequency scenarios. Unboxing, if incorrectly managed, can lead to runtime errors due to improper type casting, disrupting program execution. Additionally, these operations can increase the workload on the garbage collector, causing more frequent collection cycles and negatively impacting the application's responsiveness.



Your own company



FILL IN
SURVEY

Link to the form

- <https://forms.gle/BHnT3RatBKIPJtTP6>



Let's talk retake exam





I hope you learned:

- The importance of code readability from working on other people's code
- The importance of layer separation from trying to test things
- The importance of data access segregation from having to change it
- The importance of business logic segregation from having to change GUI/Data access or trying to test
- The importance of not murdering your friends, until the project is done
- That you can accomplish really cool stuff together.